Cyber-Resilient Secondary Frequency Control for VSG-Controlled Microgrids

Paper review, derivations, and an independent reduced-order check

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Roadmap

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- Model: swing equations on a graph
- Attack model and assumptions

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- Results, Figs. 1–4, trade-off in H
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Appendix (reference)

- Notation, Q&A, how to run the code, formula sheet

Honesty note

Slides marked **[mine]** are my own analysis or simulation, not claims of the paper.

Motivation: low inertia + a networked control layer

- **Real threat.** 2015 Ukraine grid attack [2]; a regional-supply attack on the UK is estimated at £11.6–85.5 bn of losses across sectors [3].
- **IBR-only islanded microgrids** have almost no physical inertia, so frequency moves fast after any disturbance, a cyberattack included [4,8].
- Grid-forming (GFM) IBRs set voltage and frequency; there is no physical link between frequency and power, so *droop* or *VSG* (emulated swing equation) creates one.
- **Secondary control** restores frequency and shares power, but it runs over a communication network: a natural attack surface (false data injection, FDI) [6,19].
- Prior work: mostly *detection* [16–18] (no mitigation), or resilient designs that are droop-based and ignore transient stability [20–22].

Paper's goal

Keep GFM-IBR frequencies inside a safe set under bounded FDI on the secondary-control channel,

- *without* detecting or isolating the attack,
- *with* a quantitative bound on the tolerable attack size.

Why IBR-only?

Worst case: lowest inertia, so the largest attack impact and the most stealthy-attack exposure [4]. Adding SGs only helps.

Contributions claimed by the paper

1. **Resilient controller with ancillary states.** Unlike DAI-type secondary control [13,24], it tolerates *bounded* FDI on the control inputs and does not need to identify the attack.
2. **Transient *and* steady-state resilience** for VSG-controlled microgrids, unlike [22,25]. Tolerates, rather than detects, the attack.
3. **System-theoretic certificate.** Lur'e representation + constrained input–output stability, with a performance index built on the elementwise $\mathcal{L}_\infty$ vector norm and the $\mathcal{L}_1$ system (gain-matrix) norm. A convex program returns the *maximum attack bound* $\bar{\delta}$ that keeps $\|\omega\|_{\mathcal{L}_\infty} \leq \bar{y}$ (IEEE Std 2030.8-2018).

Validation in the paper

- MATLAB: 4 GFM IBRs, full-order (LCL, inner PI loops, DC link, delays).
- Compared with DAI [24], robust [14], primal-dual [47].
- PHIL tests at LAPES (RTDS NovaCor, SEMIKRON, dSPACE).

Where the novelty sits

Not in a new plant model. It is (i) the leak/ancillary-state structure, a "retrofit" of integral control, and (ii) the *analysis machinery* that turns resilience into a computable number.

Model: swing equations on a graph (Sec. II-A)

Graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$: n IBRs, l lines, incidence matrix $E \in \mathbb{R}^{n \times l}$.

Per unit i (eq. 2):

$$\dot{\theta}_i = \omega_i, \qquad m_i \dot{\omega}_i = -d_i \omega_i + P_i^{\text{net}} - P_i + u_i$$

Lossless power flow, const. voltage (eq. 3):

$$P_i = \sum_{(i,j) \in \mathcal{E}} b_{ij} V_i V_j \sin(\theta_i - \theta_j)$$

Stacked form (eq. 4):

$$\dot{\theta} = \omega, \qquad M\dot{\omega} = -D\omega + P^{\text{net}} - E\Phi \sin(E^T \theta) + u$$

with $M{=}\mathrm{diag}(m_i)$, $D{=}\mathrm{diag}(d_i)$, $\Phi{=}\mathrm{diag}(\Phi_k)$, $\Phi_k = b_{ij} V_i V_j$ for line $k = (i, j)$.

Why $E\Phi \sin(E^T\theta)$?

$(E^T\theta)_k = \theta_i - \theta_j$ is the angle across line k; the flow $\Phi_k \sin(\cdot)$ leaves node i and enters node j. Summing over lines gives $P = E\Phi \sin(E^T\theta)$, and $\mathbf{1}^T E = 0$, so *network flows sum to zero*.

Assumption 1 (security)

There is an equilibrium θ^* with $\psi^* := E^T\theta^* \in (-\pi/2, \pi/2)^l$. It is never unique: shifting all angles by a common constant gives another one.

Modelling choices stated by the paper: lossless lines, constant voltage magnitudes, swing model valid for *system-level* frequency control; inner voltage/current loops are abstracted away.

Attack model and assumptions (Sec. II-B,C)

Actuation attack (eq. 5), additive on the secondary command:

$$M\dot{\omega} = -D\omega + P^{\text{net}} - E\Phi \sin(E^T\theta) + u + \delta_u$$

Computational attack on the controller state: δ_ξ (added in eq. 7c).

Assumption 2

$\|\delta_u\|_{\mathcal{L}_\infty} \leq \bar{\delta}_u$: "unknown but bounded", and *independent of the state trajectory*. The same is assumed for δ_ξ.

Why bounded? Unbounded or wildly out-of-range injections would be flagged by anomaly detectors, and actuators saturate. Stealthy attacks stay inside operating margins [32–34].

Control objective

Synchronism $|\omega_i - \omega_k| = 0$ under the constraint $\|\omega\|_{\mathcal{L}_\infty} \leq \bar{y}$, where $\bar{y}$ is the maximum deviation allowed by IEEE Std 2030.8-2018.

Out of scope (acknowledged)

Attacks on measurements or neighbour exchanges; unbounded attacks; state-dependent attacks. DoS and replay are only *simulated* (Case 3), not proved.

The proposed controller: an ancillary state with a leak (Sec. III-A)

Control law (eq. 6):

$$u = \beta \mathbb{K} \xi, \qquad \dot{\xi} = -\mathbb{H}\xi - \beta \mathbb{K}^T \omega$$

$\xi \in \mathbb{R}^n$ ancillary state, $\mathbb{K}$ interconnection matrix, $\mathbb{H} \succ 0$ (the **leak**), $\beta > 0$ scalar gain.

Transfer function (derive from 6b):
$\xi(s) = -(sI + \mathbb{H})^{-1} \beta \mathbb{K}^T \omega(s)$, so

$$u(s) = -\beta^2 \, \mathbb{K}(sI + \mathbb{H})^{-1} \mathbb{K}^T \, \omega(s)$$

Scalar case: $u = -\dfrac{\beta^2 K^2}{s + H}\, \omega$, a **low-pass** (lag) filter.

- $H = 0$: pure integrator, classical secondary control (DAI-type).
- $H > 0$: no integral action, so $\omega = 0$ is no longer enforced.

Topology is set by $\mathbb{K}, \mathbb{H}$

Dense $\mathbb{K}, \mathbb{H}$: centralised. $\mathbb{K}$ = Laplacian of a communication graph and $\mathbb{H}$ diagonal: *distributed* (each IBR talks to its neighbours only).

"Retrofitting"

Adds states to the standard integral loop; can be carried over to other frequency models of GFM IBRs.

[mine] DC gain of the controller

Setting $s = 0$: $u = -\dfrac{\beta^2 K^2}{H}\omega$. With the paper's numbers (β=100, K=2, H=150): $\boxed{-266.7\,\omega}$, to be compared with the physical damping $d_i = 13\,880$.

Closed loop with attacks, and its equilibrium (eq. 7)

$$\dot{\theta} = \omega \tag{7a}$$

$$M\dot{\omega} = -D\omega + P^{\mathrm{net}} - E\Phi\sin(E^T\theta) + \beta\mathbb{K}\xi + \delta_u \tag{7b}$$

$$\dot{\xi} = -\mathbb{H}\xi - \beta\mathbb{K}^T\omega + \delta_\xi \tag{7c}$$

Attack-free equilibrium. Setting $\dot{(\cdot)} = 0$ gives $\omega = 0$, then $\mathbb{H}\xi = 0 \Rightarrow \xi = 0$, then $P^{\mathrm{net}} = E\Phi\sin(E^T\theta^*)$. So $x^* = [\theta^{*T}, 0, 0]^T$.

[mine] A subtle point

x^* has $\xi^* = 0$ *only if* $P^{\mathrm{net}} \in \mathrm{Im}(E\Phi\sin(\cdot))$, i.e. no net power mismatch. After a load change the leaky controller needs $\xi \neq 0$ to push back, and then $\omega^* \neq 0$. The theory is for the attack-induced deviation from x^*, not a load step.

Lur'e representation: derivation of (8)–(9)

Let $\tilde{\theta} = \theta - \theta^*$, $q = E^T\tilde{\theta}$, $\psi^* = E^T\theta^*$. Expand the sine about ψ^*:

$$\sin(\psi^* + q) = \sin\psi^* + \operatorname{diag}(\cos\psi^*)\, q + \nu(q)$$

$$\nu(q) = \sin(\psi^* + q) - \sin\psi^* - \operatorname{diag}(\cos\psi^*)q \qquad (9d)$$

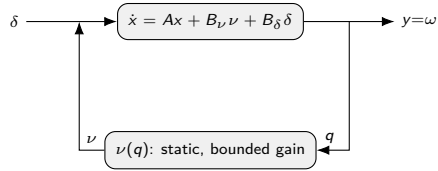
Use $P^{\text{net}} = E\Phi\sin\psi^*$ in (7b) to cancel the constant part:

$$M\dot{\omega} = -D\omega - E\Phi\operatorname{diag}(\cos\psi^*)E^T\tilde{\theta} + \beta\mathbb{K}\xi - E\Phi\,\nu(q) + \delta_u$$

With $x = [\tilde{\theta}; \omega; \xi]$: $\dot{x} = Ax + B_\nu\nu(q) + B_\delta\delta$, $y = \omega = C_y x$, $q = C_q x$, $\delta = [\delta_u; \delta_\xi]$,

$$A = \begin{bmatrix} 0 & I & 0 \\ -M^{-1}E\Phi\operatorname{diag}(\cos\psi^*)E^T & -M^{-1}D & \beta M^{-1}\mathbb{K} \\ 0 & -\beta\mathbb{K}^T & -\mathbb{H} \end{bmatrix}$$

$B_\nu = [0; -M^{-1}E\Phi; 0]$, $B_\delta = \begin{bmatrix} 0 & 0 \\ M^{-1} & 0 \\ 0 & I \end{bmatrix}$, $C_y = [0\ I\ 0]$, $C_q = [E^T\ 0\ 0]$.



Why do it?

The *linear* block carries all the dynamics and the attack; the *only* nonlinearity is a bounded static feedback, so classical input–output (small-gain) tools apply.

The nonlinear element: bounding $\nu_i(q_i)$ (Sec. III-C)

Paper: $\nu_i(q_i) = \sin(\psi_i^* + q_i) - \sin\psi_i^* - \cos(\psi_i^*)q_i$, with gain

$$\gamma_{\nu_i} = \sup_{q_i} \left| \frac{\nu_i(q_i)}{q_i} \right|, \quad \gamma_\nu = \operatorname{diag}(\gamma_{\nu_i}), \quad \|\nu\|_{\mathcal{L}_\infty} \le \gamma_\nu \, \|q\|_{\mathcal{L}_\infty} \tag{13}$$

[mine] A closed-form bound. Taylor with Lagrange remainder: for some ζ between 0 and q_i,

$$\nu_i(q_i) = -\tfrac{1}{2}\sin(\psi_i^* + \zeta)\, q_i^2 \implies \left| \frac{\nu_i}{q_i} \right| \le \frac{|q_i|}{2}$$

so, restricted to $|q_i| \le \bar{q}_i$: $\gamma_{\nu_i}(\bar{q}_i) \le \bar{q}_i/2$.

Reading it

- The nonlinear gain is *small for small angle deviations* and grows with $\bar{q}$; this is why Fig. 12(a) plots γ_ν against $\bar{q}$.
- Over the whole line, $|\nu_i/q_i| \le 2$ trivially, but that is far too conservative.
- The condition $|\psi_i^*| + \bar{q}_i \le \pi$ keeps the angle inside one sine period.

Subtle

γ_ν is valid only while $|q| \le \bar{q}$ is already known. That is circular unless a continuation (bootstrap) argument is added; see Theorem 1.

Lemma 1: the linear block is stable (Lyapunov + LaSalle)

Let $L_\psi := E\Phi\,\mathrm{diag}(\cos\psi^*)E^T$ (symmetric, weighted Laplacian since $\cos\psi^* > 0$). Candidate:

$$V(x) = \tfrac{1}{2}\tilde\theta^T L_\psi \tilde\theta + \tfrac{1}{2}\omega^T M\omega + \tfrac{1}{2}\xi^T\xi \qquad (14)$$

Derivative along $\dot x = Ax$ (the pieces):

$$\begin{aligned}\dot V &= \tilde\theta^T L_\psi\omega + \omega^T\Big(-L_\psi\tilde\theta - D\omega + \beta\mathbb{K}\xi\Big) + \xi^T\Big(-\beta\mathbb{K}^T\omega - \mathbb{H}\xi\Big)\\ &= -\omega^T D\omega - \xi^T\mathbb{H}\xi + \underbrace{\beta\omega^T\mathbb{K}\xi - \beta\xi^T\mathbb{K}^T\omega}_{=0}\end{aligned}$$

$$\boxed{\dot V = -\omega^T D\omega - \xi^T\mathbb{H}\xi \le 0} \qquad (15)$$

The coupling terms cancel because the same $\beta\mathbb{K}$ appears in u and in $\dot\xi$: a skew-symmetric (power-conserving) interconnection. LaSalle: $\dot V \equiv 0 \Rightarrow \omega = 0, \xi = 0 \Rightarrow L_\psi\tilde\theta = 0$.

[mine] Two precision remarks

1. L_ψ is *singular* ($L_\psi\mathbf{1} = 0$), so V is only positive *semi*definite in $\tilde\theta$; the claim "$V > 0\ \forall x \neq 0$" needs $\tilde\theta \perp \mathbf{1}$. LaSalle gives $L_\psi\tilde\theta = 0 \Rightarrow \tilde\theta \in \mathrm{span}(\mathbf{1})$: stability is of $(q, \omega, \xi) = 0$, i.e. *synchronisation*, not of $\tilde\theta = 0$.
2. The role of $\mathbb{H} \succ 0$: it makes ξ appear in $\dot V$. With $\mathbb{H} = 0$, $\dot V$ only controls ω; ξ is free and nothing bounds it against δ_ξ.

Gain matrices and the $\mathcal{L}_1$ bound (eq. 1, 16, 17)

Definition. For a stable $G(s)$ with impulse responses g_{ij}:
$\gamma_{G,ij} = \|G_{ij}\|_{\mathcal{L}_1} = \int |g_{ij}(\tau)|\,\mathrm{d}\tau$.

Why it bounds the peak. For $y_i = \sum_j g_{ij} * \delta_j$ (zero initial state):

$$|y_i(t)| \le \sum_j \int_0^t |g_{ij}(\tau)|\,|\delta_j(t-\tau)|\,\mathrm{d}\tau \le \sum_j \gamma_{G,ij}\,\|\delta_j\|_{\mathcal{L}_\infty}$$

$$\Rightarrow \quad \|y\|_{\mathcal{L}_\infty} \le \gamma_G\,\|\delta\|_{\mathcal{L}_\infty} \quad \text{(elementwise)}$$

The bound is *tight* for the worst-case bounded input (peak-to-peak gain).

Applied to Lur'e blocks with $y = G_{y,\delta}\delta + G_{y,\nu}\nu$ and $q = E^T \int$ y-type relation, $\gamma_{q,\cdot} = E^T_{|\cdot|}\,\gamma_{y,\cdot}$ (16):

$$\|y\|_{\mathcal{L}_\infty} \le \gamma_{y,\delta}\,\|\delta\|_{\mathcal{L}_\infty} + \gamma_{y,\nu}\,\|\nu\|_{\mathcal{L}_\infty} \qquad (17a)$$

$$\|q\|_{\mathcal{L}_\infty} \le E^T_{|\cdot|}\left(\gamma_{y,\delta}\,\|\delta\|_{\mathcal{L}_\infty} + \gamma_{y,\nu}\,\|\nu\|_{\mathcal{L}_\infty}\right) \qquad (17b)$$

Why $\mathcal{L}_\infty$ / $\mathcal{L}_1$?

Grid-code limits are *peak* limits on frequency. $\mathcal{L}_\infty$ captures the worst-case peak regardless of the energy of the disturbance; its induced system norm is exactly the $\mathcal{L}_1$ norm of the impulse response, which is what makes the bound computable.

$E^T_{|\cdot|}$

The matrix of absolute values of the entries of E^T. Needed because the bounds are *elementwise* and signs cannot cancel in a worst case.

Lemma 2: BIBO stability by a small-gain (Neumann) argument

Put $\Gamma := E_{|\cdot|}^T \gamma_{y,\nu} \gamma_\nu \in \mathbb{R}_{\geq 0}^{I \times I}$. Insert (13) into (17b):

$$\|q\|_{\mathcal{L}_\infty} \leq E_{|\cdot|}^T \gamma_{y,\delta} \|\delta\|_{\mathcal{L}_\infty} + \Gamma \|q\|_{\mathcal{L}_\infty}$$

$$\Rightarrow \; (I - \Gamma) \|q\|_{\mathcal{L}_\infty} \leq E_{|\cdot|}^T \gamma_{y,\delta} \|\delta\|_{\mathcal{L}_\infty} \qquad (18)$$

If the Perron–Frobenius eigenvalue satisfies $\lambda_{\max}(\Gamma) < 1$, the Neumann series converges:

$$(I - \Gamma)^{-1} = \sum_{k \geq 0} \Gamma^k \; \geq 0 \;\; \text{(elementwise)}$$

Multiplying (18) by a nonnegative matrix preserves the inequality:

$$\|q\|_{\mathcal{L}_\infty} \leq (I - \Gamma)^{-1} E_{|\cdot|}^T \gamma_{y,\delta} \|\delta\|_{\mathcal{L}_\infty}$$

$$\|\nu\|_{\mathcal{L}_\infty} \leq \gamma_\nu \|q\|_{\mathcal{L}_\infty}$$

$$\boxed{\|y\|_{\mathcal{L}_\infty} \leq \gamma \|\delta\|_{\mathcal{L}_\infty}}, \quad \gamma = \gamma_{y,\delta} + \gamma_{y,\nu} \gamma_\nu (I - \Gamma)^{-1} E_{|\cdot|}^T \gamma_{y,\delta} \quad (20)$$

Interpretation

γ is a finite, nonnegative *closed-loop gain matrix* from the attack δ to the frequency: a bounded input gives a bounded output (BIBO). The factor $(I - \Gamma)^{-1}$ is the amplification from the angle-feedback loop; it blows up as $\lambda_{\max}(\Gamma) \to 1$.

Typos in print (cosmetic)

Eqs. (16)–(19) mix $\gamma_{q,\nu}$, $\gamma_{y,V\nu}$ and $E_{|\cdot|}^T \gamma_{y,\nu}$ for the same object. The structure above is the consistent reading.

Theorem 1 and the resilience optimisation (Sec. IV)

Theorem 1 (paper)

If there exist $\bar{\delta}, \bar{q} \geq 0$ with

$$E_{|\cdot|}^{T}\left(\gamma_{y,\delta}\bar{\delta} + \gamma_{y,\nu}\gamma_{\nu}\bar{q}\right) \leq \bar{q}, \qquad \gamma_{y,\delta}\bar{\delta} + \gamma_{y,\nu}\gamma_{\nu}\bar{q} \leq \bar{y} \ (22)$$

then every attack with $\|\delta\|_{\mathcal{L}_\infty} \leq \bar{\delta}$ gives $\|\omega\|_{\mathcal{L}_\infty} \leq \bar{y}$.

Optimisation (23): $\max_{\bar{q}\geq 0,\ \bar{\delta}\geq 0}\ \bar{\delta}$ s.t. (22), the largest tolerable attack. Inputs: topology E, gain matrices, $\bar{y}$. Needs $|\psi_i^*| + \bar{q}_i \leq \pi$.
Role of β (Figs. 3, 12). Raising β raises the tolerable $\bar{\delta}$ (but too large a β degrades transients, more oscillation).

[mine] Gaps worth knowing

1. **Bootstrap.** (22) assumes $|q| \leq \bar{q}$ to define $\gamma_\nu(\bar{q})$. A continuation argument ($q(0) = 0$, continuity, "first time $|q| = \bar{q}$") is needed to close it; the paper says only "from Lemma 2 and (17)".
2. **Convexity.** With γ_ν fixed, (23) is a linear program. Since γ_ν depends on $\bar{q}$, convexity is asserted via [41] rather than shown.
3. **Conservatism** is admitted by the authors as future work.

The paper's evidence: test system and parameters (Sec. V, Table I)

Test microgrid. 4 GFM IBRs as VSGs, 3 lines (Fig. 4, 13); full-order model: inner PI voltage/current loops, LCL filters, DC-link dynamics, delay blocks; $\omega_0 = 2\pi 60$.

Controller: $\beta = 100$, $\mathbb{K} = 2I_4$, $\mathbb{H} = 150\, I_4$.

Parameter	Value
d_i/m_i	13 880/1000
K_{PV}/K_{IV}	0.01/1.8
K_{PC}/K_{IC}	1/33
LCL	R_f=0.05 Ω, L_f=1.5 mH, C_f=50 μF
	R_c=0.1 Ω, L_c=1.35 mH; R_g=0.03 Ω, L_g=0.35 mH
Lines	$R_{1..3}$=0.25, 0.4, 0.35 Ω; $L_{1..3}$=0.3, 1.8, 1.8 mH

[mine] What the paper does *not* give

The line susceptances Φ, the equilibrium angles ψ^*, the numerical limit $\bar{y}$, and the value of the computed $\bar{\delta}$ are not tabulated. Any reproduction of the reduced model must *assume* them.

PHIL rig (LAPES, Georgia Southern)

RTDS NovaCor real-time simulator, Spitzenberger & Spies power amplifiers (Aurora link), SEMIKRON converter with AC/DC filters, dSPACE MicroLabBox. Each GFM IBR is tested as the "hardware of interest" against the rest of the microgrid as the "model of interest".

The paper's evidence: case studies

Case	Scenario	Reported outcome
1	$\delta_u = [8\sin t,\ 7\sin(t+\frac{\pi}{2}),\ 5\sin(t+\frac{\pi}{3}),\ 3\sin(t+\frac{\pi}{2})]$ and $\delta_\xi = 1.25\,\delta_u$ (computational), $t = 5$–12 s	Frequencies stay synchronised; steady frequency slightly ($< 0.2\%$) below 1 p.u., attributed to the VSG *governor*. DAI [24], robust [14], primal-dual [47] lose synchrony and power sharing.
2	Case 1 with a 10% decrease in LCL, line, Φ, M, D	Still synchronised, power shared proportionally.
3	Replay attack: delayed replay of recorded signals (20 samples), from 5 s	Stable, zero steady-state frequency error; *no stability proof* for replay.
4	$\delta = [10\sin 10t,\ 12\sin(15t+\frac{\pi}{2}),\ 5\sin(5t+\frac{\pi}{3}),\ 6\sin(12t+\frac{\pi}{2})]$ (faster, larger)	Still resilient; faster attacks hurt more.
5	Mixed grid: an SG at IBR #2 and a GFL IBR at IBR #4, load #4 doubled for 6–9 s	Stable, acceptable performance.
PHIL	Cases 1–3 and Case 5 replicated on hardware	Close agreement with simulation.

Fig. 12: nonlinear gain vs. $\bar{q}$ and maximum attack size for $\beta = 100$; Fig. 3: maximum bound $\bar{\Delta}$ vs. β.

[mine] Independent check: what was simulated, and what was not

States $x = [\theta(4);\ \omega(4);\ \xi(4)]$: the closed loop (7) exactly, on the path graph 1–2–3–4 (3 lines).

- $M = 1000\,I$, $D = 13\,880\,I$ (Table I); $\beta = 100$, $K = 2$, $H = 150$.
- **Assumed** (not in the paper): $\Phi = 2 \times 10^4\, I_3$, $\psi^* = [0.10, -0.05, 0.08]^T$, $P^{\mathrm{net}} = E\Phi \sin\psi^*$ (makes θ^* an exact equilibrium).
- Attacks from Case 1, window 5–12 s; **sinusoidal** (paper) and **constant-bias** (for analysis): $\delta_u = [8, 7, 5, 3]$, $\delta_\xi = [10, 8.75, 6.25, 3.75]$.
- Load step: -2000 W at node 4 at $t = 2$ s, no attack.

Three controllers: primary only ($\beta = 0$), proposed ($H = 150$), pure integrator ($H = 0$, $\beta = 100$).

Why the swing level?

The theory (Lur'e form, Lemmas 1–2, Theorem 1) is derived *only* from (7). Simulating (7) tests the control law and the theory in isolation, without inner-loop tuning, PWM and filter dynamics.

What this check cannot say

Anything about LCL filters, inner loops, DC-link dynamics, delays, resistive lines, or hardware. The paper's *dramatic* baseline failures are in the full-order model; I did not reproduce those, nor the baselines [14], [24], [47] themselves ($H = 0$ is a stand-in for a pure integrator, not for DAI).

[mine] Derivation: steady state under constant bias attacks (leaky controller)

Take scalar gains M, D, K, H (as in the paper's test), constant δ_u, δ_ξ and a constant net power mismatch ΔP at a node. Assume a *synchronous* steady state $\omega_i \equiv \omega_s$, $\dot{\xi} = 0$.

(a) Sum the swing equations over nodes. $\mathbf{1}^T E = 0$ removes all network flows:

$$0 = -nD\omega_s + \beta K \sum_i \xi_i + \sum_i \delta_{u,i} + \sum_i \Delta P_i$$

(b) Each ξ equation: $0 = -H\dot{\xi}_i - \beta K \omega_s + \delta_{\xi,i} \Rightarrow \xi_i = \dfrac{\delta_{\xi,i} - \beta K \omega_s}{H}$. **(c) Insert (b) into (a):**

$$\boxed{\omega_s = \frac{H\,(\bar{\delta}_u + \overline{\Delta P}) + \beta K\, \bar{\delta}_\xi}{HD + \beta^2 K^2}}$$

(bars = mean over nodes)

Limits

- $\beta = 0$: $\omega_s = (\bar{\delta}_u + \overline{\Delta P})/D$. No ξ, so δ_ξ has no effect.
- $H \to \infty$: $\omega_s \to (\bar{\delta}_u + \overline{\Delta P})/D$ (controller switched off).
- $\beta \to \infty$: $\omega_s \to \bar{\delta}_\xi/(\beta K) \to 0$: why more β helps (Fig. 3).

Valid only for $H > 0$

For $H = 0$ each ξ_i is a *local* integrator, $\dot{\xi}_i = \delta_{\xi,i} - \beta K \omega_i$, which pins $\omega_i \to \delta_{\xi,i}/(\beta K)$ node by node. A synchronous steady state generally does not exist (next slides).

[mine] Numbers: closed form vs. simulation (constant bias, $t \in [11.5, 12]$ s)

$\max_i \lvert\omega_i\rvert$ (rad/s)	Primary only	Proposed (H=150)	Pure integrator (H=0)
δ_u only	4.1×10^{-4}	4.1×10^{-4}	≈ 0 (rejected)
δ_ξ only	0 (no ξ)	6.8×10^{-4}	4.7×10^{-2}
both	4.1×10^{-4}	1.09×10^{-3}	4.7×10^{-2}
Load -2000 W, final ω	-3.60×10^{-2}	-3.53×10^{-2}	≈ 0 (restored)
Sinusoidal attack (Case 1), peak	5.6×10^{-4}	1.46×10^{-3}	4.0×10^{-2}

Closed form (c) for $H = 150$: denominator $HD + \beta^2 K^2 = 2.122 \times 10^6$.
δ_u only: $150 \cdot 5.75/2.122 \times 10^6 = 4.06 \times 10^{-4}$
δ_ξ only: $200 \cdot 7.1875/2.122 \times 10^6 = 6.77 \times 10^{-4}$
both: $2300/2.122 \times 10^6 = 1.084 \times 10^{-3}$
load: $150 \cdot (-500)/2.122 \times 10^6 = -3.53 \times 10^{-2}$

These match a separate Python/SciPy re-implementation of the same ODE; check them against your own Octave console output (the first block of `jamali_vsg_demo.m` prints exactly the constant-bias rows). The sinusoidal-peak row is a numerical value, not a closed form.

Reading the H=0 column

4.7×10^{-2} is node 1 (individual nodes sit at 4.7, 4.2, 3.2, 2.3×10^{-2}, not equal): synchrony is lost, as the paper describes for its baselines. Compare the uncoupled pin $\delta_{\xi,i}/(\beta K) = 5.0, 4.4, 3.1, 1.9 \times 10^{-2}$.

[mine] What the leak does: dynamics and effective droop

Common-mode characteristic polynomial (scalar gains, network flow terms cancel in the mean):

$$Ms^2 + (D + MH)s + (DH + \beta^2 K^2) = 0$$

	polynomial / M	poles (s^{-1})
H=0	$s^2 + 13.88s + 40$	$-4.1,\ -9.8$
H=150	$s^2 + 163.9s + 2122$	$-14.2,\ -149.7$
primary	$s + 13.88$	-13.88

Attack gain at 1 rad/s ($\delta_\xi \to \omega$, common mode):
$|G| = \dfrac{\beta K}{|\,(Ms{+}D)(s{+}H) + \beta^2 K^2|}$ gives 4.8×10^{-3} (H=0) vs. 9.4×10^{-5} (H=150): about 51× attenuation per unit of common-mode δ_ξ.

The leak = a small extra droop

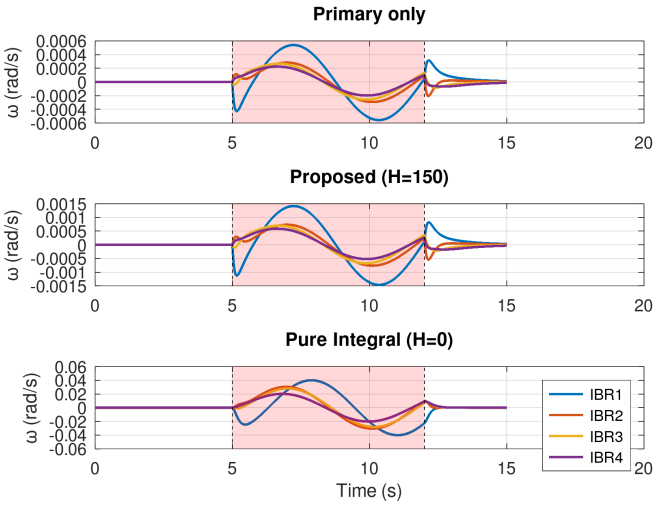
Below the corner $H = 150$ rad/s, far above the electromechanical band (∼4–14 s^{-1}), the controller behaves as a *static* gain $u = -(\beta^2 K^2/H)\omega$:

$$D_{\text{eff}} = D + \frac{\beta^2 K^2}{H} = 13\,880 + 266.7\ (+1.9\%)$$

Consequence

At $H = 150$ the controller delivers resilience (about 50× vs. H=0 on δ_ξ) and gives back only 1.9% of the load-step frequency error relative to primary control (3.53 vs. 3.60×10^{-2} rad/s). Almost all of the "restoration" has been traded away. The paper's own claim that the leak causes no problem rests on the governor explanation for the 0.2% offset in the *full* model, which I cannot test at swing level.

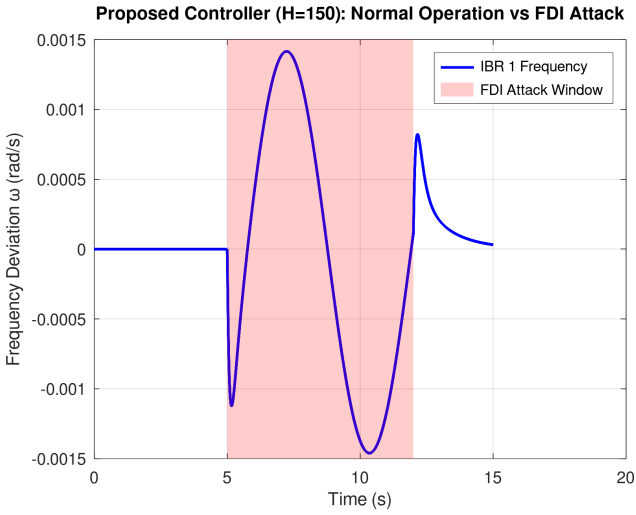
[mine] Fig. 1: three controllers under the paper's sinusoidal attack



Shaded: attack window 5–12 s.

- Numerical peaks of $|\omega|$ during the attack: primary 5.6×10^{-4}, proposed 1.46×10^{-3}, $H{=}0$ 4.0×10^{-2} rad/s, i.e. the leaky controller is about $27\times$ better than the pure integrator and about $2.6\times$ *worse* than having no secondary loop at all.
- $H = 0$: individual frequencies split during the attack; after $t = 12$ s they return to 0, but ξ and the angles settle at *new* offsets (the integrator never forgets).
- $H > 0$: sync is kept, small ripple at the attack frequency.

[mine] Fig. 2: proposed controller, normal → attack → normal

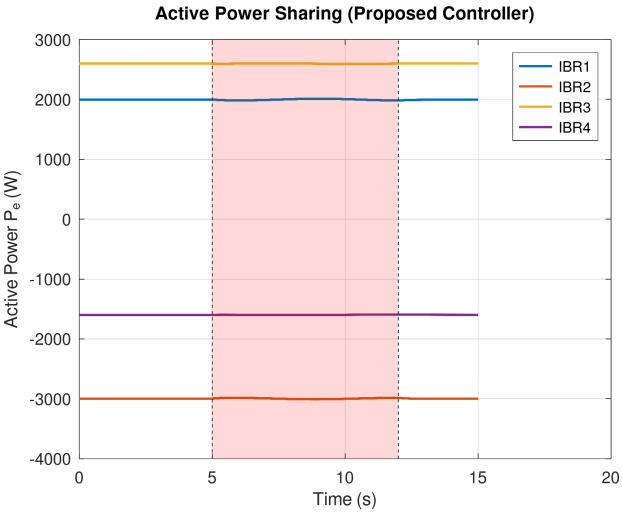


- 0–5 s: at the equilibrium ($\omega = 0$). 5–12 s: bounded response, peak $\sim 10^{-3}$ rad/s, i.e. $\sim 4 \times 10^{-6}$ of $\omega_0 = 377$ rad/s.
- After 12 s: returns to 0 within about a second (poles at -14 and $-150\,\mathrm{s}^{-1}$), then stays: the leak forgets the attack.
- Theory check: Lemma 1 ($\dot{V} \leq 0$) plus bounded inputs gives a bounded response, as seen.

Careful

Peak $|\omega| \ll \bar{y}$ here does not by itself validate Theorem 1's *bound*; that needs γ matrices, which I did not compute.

[mine] Fig. 3: network power injections during the attack



- Plotted: $P_e = E\Phi \sin(E^T\theta)$, the net *network injection* at each IBR.
- This is **not** a proportional-power-sharing metric: that needs the droop/VSG ratios, and the paper shows it only in its full-order Figs. 5–11.
- Use it to show that line flows stay far from the $\pm\Phi$ limit (2×10^4 W) during the attack, i.e. Assumption 1 is not threatened.

Do not claim

"Power sharing remains proportional" from this plot.

[mine] Why the pure integrator is the real hazard: differential-mode drift

With $H = 0$ and diagonal K: $\dot{\xi}_i = \delta_{\xi,i} - \beta K \omega_i$. A synchronous steady state would need $\omega_i = \omega_s$ for all i, hence $\dot{\xi}_i = \delta_{\xi,i} - \beta K \omega_s$, which is nonzero for every node whose bias differs from the mean:

$$\delta_{\xi,i} - \bar{\delta}_\xi = (+2.81,\ +1.56,\ -0.94,\ -3.44)$$

$$\dot{u}_i = \beta K \dot{\xi}_i = (+563,\ +313,\ -188,\ -688)\ \text{W/s}$$

So the u_i would ramp linearly (wind-up). The system compromises: ω_i separate (values above), the angles slip at a constant rate instead:

	q_1	q_2	q_3 (rad)
$t = 12$ s (end of real attack)	0.14	0.02	0.14
$t = 80$ s (attack continued)	0.52	0.69	0.75
$t = 200$ s	0.96	2.41	2.09

Numerical, $H = 0$ and bias attack held on. For $H = 150$ the angles stay at their equilibrium values $(0.10, -0.05, 0.08)$.

Reading it

- In the 7 s of the paper's scenario the damage is a frequency split of about 2.4×10^{-2} rad/s, small in absolute terms.
- The structural issue is that the angle differences have *no restoring equilibrium*: held long enough they pass $\pm\pi/2$ (Assumption 1) and synchronism is lost.
- With the leak, $\xi_i = (\delta_{\xi,i} - \beta K \omega_s)/H$ is bounded, so u_i is bounded, so the angles stay put.

Optional extra experiment

In `rhs_func.m` change `t <= 12` to `t <= 300`; in `run_sim.m` change the 12 in `breaks`; use the `'bias'` attack and `Tend = 300`. Compare $H = 0$ vs. 150.

[mine] The resilience vs. restoration trade-off in H (closed form)

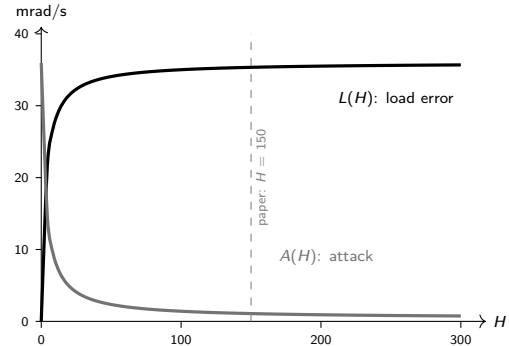
From the boxed formula (both attacks, load step $-500\,\mathrm{W}$ per node on average):

$$\text{attack: } A(H) = \frac{5.75\,H + 1437.5}{13.88\,H + 40}\ \text{mrad/s}$$

$$\text{load: } L(H) = \frac{500\,H}{13.88\,H + 40}\ \text{mrad/s}$$

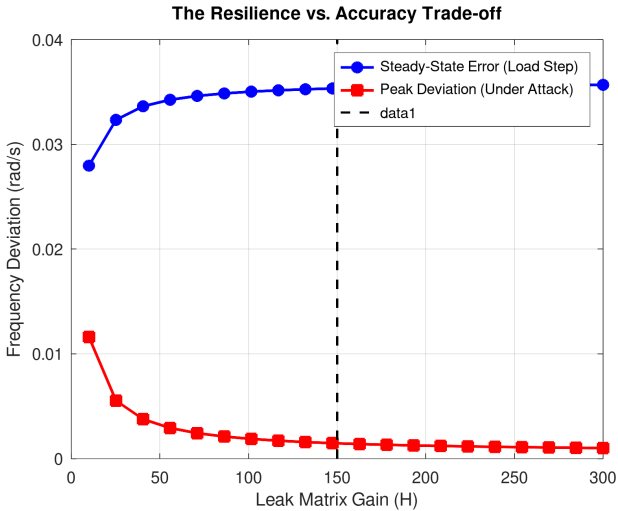
- A falls monotonically from 35.9 at $H \to 0^+$ to 0.4143 as $H \to \infty$; L rises from 0 to 36.0.
- They cross at $H^\dagger = \beta K \bar{\delta}_\xi / (|\overline{\Delta P}| - \bar{\delta}_u) \approx 2.9$, not near the paper's $H = 150$.
- At $H = 150$: $L = 35.3$ (98% of the primary-only error), $A = 1.08$.

Numerical sweep (sinusoidal attack, node-1 peak) crosses between $H = 3$ and 10 as well.



Analytic curves, constant bias attack. Compare with Octave Fig. 4 (next slide).

[mine] Fig. 4: numerical H-sweep from the Octave script



- Sweep of $H \in [10, 300]$: load-step error (blue) rises and saturates near 3.5×10^{-2}; sinusoidal-attack peak (red) falls.
- Numerical values (sin. attack, node 1): H=10: 1.2×10^{-2}; 50: 3.2×10^{-3}; 150: 1.5×10^{-3}; 300: 1.0×10^{-3} rad/s.
- $H = 150$ sits on the *flat* part: the load error is already saturated, while the attack peak is still falling slowly.

Do not claim

A "scissors crossing at the paper's $H = 150$". The crossing is at small H (single digits).

Strengths

Conceptual

- **Detection-free resilience.** Bounded attacks are tolerated by structure (a leaky state), not by recognising them.
- **Resilience as a number.** $\bar{\delta}$ from a small optimisation problem, in the same units as the attack, tied to a grid-code limit $\bar{y}$.
- **Clean proof skeleton.** $\dot{V} = -\omega^T D\omega - \xi^T \mathbb{H}\xi$ (the skew coupling cancels) + $\mathcal{L}_1$ gains + small gain.

Practical

- Retrofit of integral control; distributed implementation if $\mathbb{K}$ is a Laplacian and $\mathbb{H}$ diagonal.
- Evidence beyond ideal simulation: parameter perturbation, replay, mixed SG/GFL grid, PHIL hardware tests.
- My reduced-order check agrees with the mechanism: a leak bounds the effect of a bias on ξ by orders of magnitude (about 40–70$\times$ in the constant-bias δ_ξ rows).

Limitations and critique

1. Restoration is approximate by design. $\omega_s = 0$ is lost under any persistent mismatch (formula (c)). At the tested tuning the leak returns only $\approx 1.9\%$ of the load error. The paper attributes the $< 0.2\%$ offset to the VSG governor; the leak plausibly contributes too, but this cannot be separated at swing level.

2. Baselines vs. attack channel. The paper does not say how the computational-attack analogue δ_ξ is applied to DAI/robust/primal-dual, so the comparison may not be like for like. In my reduced model, *no* secondary loop beats the leaky one under bias attack (4.1 vs. 10.9×10^{-4}): the controller itself adds attack surface.

3. Scope of the attack model. Additive, bounded, state-independent, on u and ξ only. Measurement/neighbour-exchange, DoS, replay, state-dependent (adaptive) attacks are excluded; Case 3 is simulation only.

4. Proof details. V is only semidefinite (angle shift invariance); Theorem 1 needs a continuation argument to justify $\gamma_\nu(\bar{q})$; convexity of (23) is asserted via [41]; typos in (16)–(19).

5. Reproducibility. Φ, ψ^*, $\bar{y}$, and the actual $\bar{\delta}$ are not reported; no tuning guidance for $(\beta, \mathbb{H})$ beyond "large β gives oscillation".

6. Conservatism (admitted): the $\mathcal{L}_1$ bound with elementwise worst-case combining is likely loose.

7. Equilibrium assumption. $\xi^* = 0$ presumes no net mismatch; load changes shift the leaky controller to $\xi^* \neq 0,\ \omega^* \neq 0$, outside the theory.

Conclusion

Take-aways

1. The paper makes secondary frequency control *tolerant* to bounded FDI by adding a leaky ancillary state, and certifies it with a Lur'e-form, $\mathcal{L}_\infty/\mathcal{L}_1$ small-gain analysis and a convex "maximum attack" program.
2. Mechanism, in one line: $u = -\beta^2 K(sI + H)^{-1} K^T \omega$. The leak bounds the controller's response to injected bias; the price is exact restoration.
3. My reduced model confirms the mechanism and shows the cost: $\approx$ 40–70$\times$ less deviation than a pure integrator under constant bias, but only $\approx 2\%$ restoration versus primary control at $H = 150$.

Open questions

- How should (β, H) be chosen for a target mix of attack tolerance and restoration?
- Can the leak be made adaptive, off when no attack is present?
- Extension of the guarantee to measurement attacks, DoS, state-dependent attacks.
- Less conservative certificates than $\mathcal{L}_1$ gains.

Thank you. Questions?

Appendix A: notation and symbol glossary

θ, ω	angle, frequency deviation of IBRs
M, D	inertia, damping (diagonal)
P^{net}	nominal power injection
E	node–line incidence matrix ($n \times l$)
Φ	line weights $b_{ij} V_i V_j$ (diagonal)
ψ^*	equilibrium angle differences $E^T \theta^*$
q	$E^T(\theta - \theta^*)$, angle-difference error
$\nu(q)$	nonlinear remainder of sin
ξ	ancillary controller state
$\beta, \mathbb{K}, \mathbb{H}$	gain, interconnection matrix, leak

δ_u, δ_ξ	actuation, computational FDI
$\bar{\delta}, \bar{y}, \bar{q}$	bounds on attack, frequency, angle error
$\|\cdot\|_{\mathcal{L}_\infty}$	elementwise sup-norm (vector)
γ_G	gain matrix, entries $\left\| G_{ij} \right\|_{\mathcal{L}_1}$
γ_ν	diagonal gain of ν_i
$E^T_{\lvert\cdot\rvert}$	E^T with absolute-value entries
$\lambda_{\max}$	Perron–Frobenius eigenvalue
BIBO	bounded-input bounded-output
VSG, GFM, GFL	virtual synchronous gen., grid-forming, grid-following
DAI	distributed averaging integral control
PHIL	power hardware-in-the-loop

Appendix B: anticipated questions (1/3)

Q1. Why simulate only the swing level, not LCL filters, inner loops and DC link like the paper?
The paper's theory (Lur'e form, Lemmas 1–2, Theorem 1) is derived from (7) alone. Simulating (7) tests the control law and the theory without confounding inner-loop tuning. The price: I make no claim about full-order behaviour or about the baselines [14], [24], [47].

Q2. Does the pure integrator "wind up to infinity"?
Not in the frequency: with local integrators each ω_i is pinned near $\delta_{\xi,i}/(\beta K)$ (about 4.7×10^{-2} rad/s max). What grows is the *angle drift*: the angle differences have no equilibrium and slip at a constant rate (see differential-mode slide). In the 7 s window it is a frequency split; sustained, it would cost synchronism.

Q3. Why is the leak $\mathbb{H} \succ 0$ needed in the proof?
$\dot{V} = -\omega^T D\omega - \xi^T \mathbb{H}\xi$. With $\mathbb{H} = 0$ the ξ term disappears from $\dot{V}$: only ω is controlled, ξ is free, and a bias on ξ has no restoring force. With $\mathbb{H} \succ 0$, ξ is exponentially damped, which gives finite gains γ.

Q4. Why $\mathcal{L}_\infty$ and $\mathcal{L}_1$ rather than $\mathcal{L}_2/H_\infty$?
Grid codes limit the *peak* frequency. The $\mathcal{L}_\infty$-induced norm of an LTI system is the $\mathcal{L}_1$ norm of its impulse response (peak-to-peak gain), so the bound is both tight and computable elementwise.

Q5. The leaky controller never restores frequency to exactly 0; why call it secondary control?
Correct, and my formula shows it: $\omega_s = [H(\bar{\delta}_u + \overline{\Delta P}) + \beta K \bar{\delta}_\xi]/(HD + \beta^2 K^2)$. It is a deliberate trade: exact integral action is exactly the component a bias attack exploits. At $H = 150$ the restoration left is small ($\approx$ 2% of the load error).

Appendix B: anticipated questions (2/3)

Q6. In your table, primary-only beats the proposed controller under attack. Doesn't that defeat the point?
Yes in magnitude: a loop that does not exist cannot be attacked (4.1×10^{-4} vs. 10.9×10^{-4} rad/s). But primary-only restores nothing (load error 3.60×10^{-2}). The paper's claim is relative to *integral* secondary control, and my numbers support that ($\approx$ 40–70$\times$ better on δ_ξ).

Q7. What if the attack exceeds the computed bound $\bar{\delta}$?
The certificate is lost. Physically, the linear part stays BIBO stable but $|q|$ can exceed $\pi/2$ (Assumption 1), where the sine nonlinearity stops being a stabilising restoring force: synchronism can be lost. Unbounded attacks are excluded by Assumption 2.

Q8. What does β do?
From my formula the attack-induced deviation scales as $\sim \bar{\delta}_\xi/(\beta K)$ for large β, consistent with Fig. 3: larger β tolerates larger attacks. The paper warns that too large a β worsens oscillation (the common-mode poles move: $s^2 + (D/M{+}H)s + (DH{+}\beta^2K^2)/M$).

Q9. Is (23) really convex?
With γ_ν fixed it is linear in $(\bar{\delta}, \bar{q})$. The dependence $\gamma_\nu(\bar{q})$ is handled by citing [41] and the bound $|\psi_i^*| + \bar{q}_i \leq \pi$; the paper does not prove it in situ.

Q10. How do you know your simulation is right?
The leaky-controller constant-bias numbers agree with the closed-form steady-state formula to three digits, and with a separate Python implementation of the same ODE. The $H = 0$ numbers cannot be checked in closed form (no synchronous steady state), only numerically.

Appendix B: anticipated questions (3/3)

Q11. Why is the paper's Lyapunov function acceptable if it is not positive definite?
L_ψ has the kernel $\mathrm{span}(\mathbf{1})$. Define V on (q, ω, ξ) or restrict to $\tilde{\theta} \perp \mathbf{1}$; LaSalle then yields convergence to synchronisation. It is a presentational imprecision, not an error in substance.

Q12. Where does the paper's 0.2% frequency offset come from?
The paper: the VSG governor/power-sharing characteristic. My view: in the *leaky* controller a persistent mismatch always leaves $\omega_s \neq 0$ (formula (c)); the governor and leak effects cannot be separated without the full-order model.

Q13. Distributed implementation?
Take $\mathbb{K}$ = Laplacian of the communication graph, $\mathbb{H}$ diagonal: each IBR needs only neighbour values. Note: with $\mathbb{K}$ *diagonal* (as in the paper's test, $2I_4$) the controller is purely local.

Q14. Replay and DoS?
Not covered by theory; Case 3 is a simulation of replayed signals (20-sample offset) with stable behaviour. The paper itself says no scheme resists all attacks (Remark 3).

Q15. Why does the baseline fail so dramatically in the paper but not in your model?
I do not know; I did not reproduce the baselines. A plausible factor is the full-order dynamics and delays; another is the attack channel used for the baselines (see critique point 2). Say so rather than guess.

Appendix C: running the code (reference)

Files in `~/projects/jamali-vsg/`

- `jamali_vsg_demo.m`: main script: constant-bias table, load step, Figs. 1–4, saves PNGs.
- `run_sim.m`: segmented ode45 integration with breakpoints $[0, 2, 5, 12, T_{\mathrm{end}}]$ (events at load step and attack on/off).
- `rhs_func.m`: right-hand side of (7) plus attack and load-step logic.

`octave -no-gui jamali_vsg_demo.m`
Local functions at the end of a script file failed on this Octave version, hence the three files. `yyaxis` does not exist in Octave: Fig. 4 uses one axis.

Mapping code ↔ paper

Code	Paper
`M,D`	M, D (Table I)
`E, Phi`	E, Φ (Phi assumed)
`beta, H`	β, $\mathbb{H}$ (6)
`K = 2*eye(4)`	$\mathbb{K} = 2I_4$
`a(1:4), a(5:8)`	δ_u, δ_ξ (7b,7c)
`dx(1:4)`	$\dot{\omega}$ in (7b)
`dxi`	$\dot{\xi}$ in (7c)

Compile this deck: `pdflatex seminar.tex` (twice). Figures are read from the same folder.

Appendix D: formula sheet

Plant and controller

$$\dot{\theta} = \omega$$
$$M\dot{\omega} = -D\omega + P^{\mathrm{net}} - E\Phi\sin(E^T\theta) + \beta\mathbb{K}\xi + \delta_u$$
$$\dot{\xi} = -\mathbb{H}\xi - \beta\mathbb{K}^T\omega + \delta_\xi$$

Lur'e form: $\dot{x} = Ax + B_\nu\nu(q) + B_\delta\delta,\ y = C_y x,\ q = C_q x$
Lyapunov: $V = \frac{1}{2}\tilde{\theta}^T L_\psi\tilde{\theta} + \frac{1}{2}\omega^T M\omega + \frac{1}{2}\xi^T\xi,\ \dot{V} = -\omega^T D\omega - \xi^T\mathbb{H}\xi$
Nonlinear gain: $|\nu_i/q_i| \le |q_i|/2$

Small gain / bound

$$\Gamma = E_{|\cdot|}^T\gamma_{y,\nu}\gamma_\nu, \quad \lambda_{\max}(\Gamma) < 1$$
$$\gamma = \gamma_{y,\delta} + \gamma_{y,\nu}\gamma_\nu(I-\Gamma)^{-1}E_{|\cdot|}^T\gamma_{y,\delta}$$
$$\|y\|_{\mathcal{L}_\infty} \le \gamma\,\|\delta\|_{\mathcal{L}_\infty}$$

Max attack: $\max\bar{\delta}$ s.t. $E_{|\cdot|}^T(\gamma_{y,\delta}\bar{\delta} + \gamma_{y,\nu}\gamma_\nu\bar{q}) \le \bar{q}$, $\gamma_{y,\delta}\bar{\delta} + \gamma_{y,\nu}\gamma_\nu\bar{q} \le \bar{y}$
[mine] Steady state (leaky, scalar gains):

$$\omega_s = \frac{H(\bar{\delta}_u + \overline{\Delta P}) + \beta K\bar{\delta}_\xi}{HD + \beta^2K^2}$$

Effective droop: $D_{\mathrm{eff}} = D + \beta^2K^2/H$. Common-mode poles: $Ms^2 + (D{+}MH)s + DH + \beta^2K^2 = 0$.